

Question-Answering Systems: Challenges and Approaches

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Abstract

Question Answering Systems (QAS) are gaining popularity in the advancement of AI, and enabling efficient information retrieval across various domains. This paper explores the challenges and approaches associated with QAS, focusing on both extractive and generative methodologies. We delve into key issues such as lexical and syntactic variations, multi-hop reasoning, ambiguity, model hallucinations and provide use case scenarios for different types of language models. The paper also presents SOTA developments in models like BERT and ChatGPT, illustrating their capabilities and limitations in QAS applications. Through this comprehensive analysis, we aim to explore the current state of QAS technology and mention areas of future research.

1 Introduction

The time we revolutionized how we interact with computers has arrived. Amount of resources it takes surfing on the Web to achieve information, is getting shorter by day with the emergence of new Natural Language Processing(NLP) techniques. One such technology is called Question Answering Systems(QAS), that focus on making the information retrieval as intuitive as possible, allowing users to pose their questions in Natural Language and receive concise answers. Differently from the process of using search engines by coming up with the best keywords and going through numerous links, Question Answering Systems get closer to understanding what we seek from prompts, bridging the gap between human and machine understanding.

Some key features of QAS include in which ways users can inquire information. For example, the user can seek Factual Information, to receive specific data points such as “What is the capital of France?”. The user can just as well involve images, with Visual Questions to ask “Which vehicle is shown in this picture?”. The questions can

also request Open-Goal Oriented Questions such as “Tell me about photosynthesis” to get descriptive answers.

Responses differ in the nature that they are produced. The system can approach in an Extractive manner and directly retrieve a passage that matches the query, or synthesize the answer in a Generative manner to merge information retrieved from multiple sources into a contextually accurate answer, demonstrating a deep understanding of Natural Language. These approaches will be explored more in depth in the following sections of this paper.

2 History of Question Answering Systems

Conversational QAS were first developed in the 60s. Weizenbaum developed a system called ELIZA, which simulates a psychotherapist ([Weizenbaum, 1966](#)),. Following this feat was BASEBALL ([Green Jr et al., 1961](#)), specializing in answering questions about various baseball leagues. The research done by Green B.F. et al. shows the need to disambiguate the user’s question and problem of multiple implied contexts, which are relevant challenges even today.

In the 1990s, the START ([Katz, 1988](#)) system introduced the use of ternary expressions, which are a set of nested subject-relation-object triples, and how effective they are in the context of QAS to handle more complex relationships between entities and showed that the development of how we represent data is pivotal in QAS technology ([Voorhees and Tice, 2000](#)).

In the 2000s, IBM’s Watson won the Jeopardy! game in a real-time competition with previous champions. This feat demonstrated the potential of the QAS field as the game inherently requires a fast understanding of various real-world concepts and human-like commonsense reasoning (?). In this era, the Text Retrieval Conference (TREC QA) played a crucial role in developing benchmarks to

advance and test QAS technologies.

The rise of Deep Learning in the 2010s accelerated the advancement of QAS by introducing elegant methods of Machine Learning and freed researchers from tedious feature engineering, as well as promising a bigger capacity to capture long-term dependencies and uncovering hidden patterns in textual data (Hao et al., 2022). Models like BERT and GPT-2 showed great potential in the effectiveness of Neural Networks in contextual understanding.

Currently in the 2020s, the usage of QAS has become more accessible and more accurate than ever before with the emergence of SOTA technologies achieving new milestones such as PaLM, the first model to exceed a “passing” score in US Medical Licensing Examination (USMLE) style questions with a score of 67.2% on the MedQA dataset (Singhal et al., 2023). Github’s Codex also shows great potential in generating code snippets comparable to experienced programmers (Wermelinger, 2023), and Alibaba’s AliceMind shows high performance on the Chinese Language Understanding Evaluation (CLUE) benchmark, especially in Visual Question answering (Yan et al., 2021).

3 QAS Taxonomy

In this section, we will be going over the features that differentiate QAS and provide taxonomic information (Abdel-Nabi et al., 2023). The reader should note that there are multiple ways to interpret QAS taxonomy and this version is deemed enough for the scope of this paper.

Context Type refers to the length and extent of the input provided by the user to the QAS. It can be simple factual statements, short paragraphs called passages, complete documents, or multiple documents aggregating the information from several documents.

Context Source can vary, starting from exams that can provide question-answer pairs, news articles about recent events, general web-pages, or most commonly, Wikipedia articles with encyclopedia-like entries and a graph-like structure of relevant articles linking to each other, revealing relevancy between entities, usually equipped with search engines’ logs.

Reasoning Type can be categorized by the number of passes over the text to find the correct answer. Single-hop, referring to answers derived from a single piece of information, or multi-hop, referring

to the internal reasoning mechanism of the QAS piecing together multiple sources of information in a logical way to conduct the answer. Multi-hop reasoning can occur within a single document when the necessary information is scattered within the source material, or multiple documents, where QAS has to connect the relevant information from a multitude of documents, proving to be the hardest task within this section to implement. For instance, answering the question "Where was the scientist who discovered penicillin born?" might involve first identifying the scientist (Alexander Fleming) and then finding his birthplace (Lochfield, Scotland).

Domain Type refers to the scope of information the system is designed to handle. Closed-domain or Domain-dependent QAS restrict the Knowledge Base and contains a specific domain terminology and vocabulary. The answers extracted from such a system are prone to be more exact and consistent, used in environments such as medical records, legal documents, or company FAQs, relying on a more focused dataset. Open-domain or Domain-independent QAS leverages general world knowledge and can operate on any provided knowledge on the dataset, without requiring a specific set of terms. Even though these systems are versatile and more suited to daily use, they require the involvement of huge amounts of data to be processed.

Answer Format reflects how the answer to the user’s question is fetched. Extractive methods will provide spans of text that is the target of the question, meanwhile Retrieval& Ranking methods will collect multiple answers from documents and assign the most probable answer as the result of the query. Abstractive/Generative methods will construct a descriptive answer in Natural Language from scratch (Rajpurkar et al., 2016).

4 Extractive QAS

Extractive Question Answering Systems are designed to identify and extract exact text spans from a source document that answer a given query, utilizing Reading Comprehension techniques. This methodology is a supervised learning problem that depends mostly on the training data (Rajpurkar et al., 2016).

4.1 General Architecture of Extractive QAS

The general architecture can be summarized in three parts:

Input embeddings: The extractive QAS begins by tokenizing the words in the context (passage or document) into smaller units that carry the semantic meaning. Both the context, the scope or domain, and the question will then be transformed into numerical vector representations by making use of word embedding techniques. The context and the question is fed to the Feature Extraction module (Wang et al., 2022).

Feature Extraction: This module extracts pertinent features from both the context and the question, highlighting possible answer regions from the context, and identifying the specific information that is sought from the question. Most commonly, Recurrent Neural Networks(RNNs), Convolutional Neural Networks(CNNS), and Transformer structure based on multi-head self-attentive mechanisms are used to achieve these results. These features will then be fed to the Context-Question Interaction Module (Wang et al., 2022).

Context-Question Interaction: This module integrates the cumulative information obtained from above modules and compares results based on their relevance and alignment. The model predicts the start and end positions of the answer span within the text using a probability distribution over all possible positions. The most likely passage from the text will be highlighted as the possible answer and will be passed down to the Context-Question module again, to dig deeper into the relationship between the article and the question. This interaction can be executed several times iteratively to simulate the human behavior of repeated reading. Most commonly, unidirectional or bidirectional attention mechanisms are used to capture the most likely answer. Evaluation metrics such as Exact Match (EM) and F1 score measure the model's precision and recall in identifying correct answer spans (Wang et al., 2022).

4.2 Extractive Question Answering System

Example: BERT

Introduced in 2018, BERT (Bidirectional Encoder Representations from Transformers) is a state-of-the-art pre-trained transformer model that excels in various Natural Language Processing tasks. BERT skillfully captures various quirks of Natural Language such as hierarchical relationships, high-level (indirect, implied or hidden) semantics, and syntactic inference (analyzing the grammatical structure of sentences to understand the relationships

between phrases).

Notable characteristics of the BERT architecture involve Bidirectional Reading to concurrently consider both left and right context for each token, Masked Language Model(MLM), where random tokens in the input are masked, and the model learns to predict these masked tokens based on the surrounding context, reminiscent of the way humans would learn a foreign language by reading excerpts from their target language and trying to infer the meaning of unknown words without looking at the dictionary. The input presented to BERT is a concatenated sequence of the question and the context, separated by special tokens ([CLS] and [SEP]) (Nassiri and Akhloufi, 2023).

One other unique feature of BERT is the ability to use Fine-tuning on task-specific datasets, such as SQuAD (Stanford Question Answering Dataset), where it learns to predict the start and end offsets of the answer span within a given passage. During the fine-tuning phase of the training, BERT adapts its parameters to specific QA tasks by minimizing a cross-entropy loss function over the predicted answer span locations. It is worth keeping in mind that the goal format of an extractive QAS is the starting and ending word indexes of the target answer.

Unlike traditional word embeddings (like Word2Vec or GloVe), which assign a single numerical vector to a word regardless of context opening up the possibility of missing out on the nuances of Natural Language, BERT generates contextual vector embeddings. Meaning, the same word would be encoded into a different vector if the text is even slightly changed. These embeddings take the surrounding words into consideration, allowing BERT to distinguish between different meanings of a word based on a possibly ambiguous context, such as "bank" as a financial institution or "bank" of a river.

BERT is famously based on the Transformer architecture, collaboratively with the self-attention mechanisms to capture dependencies and long-distance references that may possibly be scattered all over the course of multiple documents. Identifying long-term dependencies allows BERT to not discard the information that appears earlier or later in the text, lessening the likelihood of missing out on important content. Moreover, the architecture makes BERT the perfect candidate for parallelized processing, enabling the computation

to be distributed over a number of hosts to make BERT scalable and fast. Models based on BERT should be noted to perform better on content that is similar to the original training data, such as questions and answers that match the exact wording on Wikipedia.

4.3 Challenges of Extractive Question-Answering Systems

Since 2018, it is fair to state that BERT and other similarly built extractive models have pushed boundaries of NLP. In this section, we will explore the challenges that are yet to be completely overcome by extractive QAS (Rajpurkar et al., 2016).

Lexical Variation - Synonymy: Since extractive QAS often rely on finding exact (or partial) word matches within the training corpus of the model and the question, synonymy poses a significant threat to the accuracy of retrieved results. Synonymy occurs when different words or phrases convey the same meaning. For instance, the query “What is the capital of France?” and “Which city serves as the capital of France?” both seek the same information, yet the terms “capital” and “city” introduce an element of confusion that may be easily discarded by the human eye.

Lexical Variation - World Knowledge: Major correspondences between question and answer pairs may require seemingly obvious world knowledge to decipher. For example, the question “Which governing bodies have veto power?” may pose confusion for the QAS, as what can be considered as a “body” or an abstract concept such as “governing” being matched to a physical body needs general world knowledge to disambiguate the meaning.

Syntactic Variation: A single piece of information can be conveyed through multiple syntactic forms, complicating the extraction process. After the question is paraphrased, the syntactic form of the question may not match the wording in the text possibly containing the answer, raising the likelihood of overlooking a seemingly obvious match. For instance, the question “How is the process of photosynthesis described?” might be rephrased as “What is the description of photosynthesis?”. Effective handling of syntactic variation requires robust parsing techniques and the ability to recognize and adapt to various syntactic patterns, which now remains as an area of future work in the research field.

Multiple Sentence Reasoning: Extractive systems, which typically focus on identifying the most relevant sentence or phrase, may find it difficult to aggregate and integrate information across multiple sentences. Many questions posed to the QAS are likely to request information that can be synthesized with high-level reasoning to unveil the dependencies between entities that can be discovered in a vast selection of knowledge bases over different domains.

Ambiguity: Query ambiguity occurs when a question can be interpreted in multiple ways. As an example, the question “What are the advantages?” could refer to advantages in various contexts, such as technology, education, or health. Techniques for addressing ambiguity often involve incorporating disambiguation strategies, context modeling, and advanced NLP techniques to better understand and resolve ambiguities. It is worth considering that the question may not have a unique answer at all.

4.4 Comparison with Human Performance

While extractive QAS have advanced significantly over the last years, due to the reasons listed above, and more, they continue to face struggles where human performance easily outshines the results of these language models. One such study starts by defining syntactic divergence as the minimum edit distance over all possible anchors, or in other terms, keywords. In this study, human performance kept an unwavering score even with the number of required edits necessary to get to the original text has increased over time, meanwhile the extractive QAS model has decreased in F1 Score almost immediately after derailing from the source material. Researchers also noted that the most human “errors” were recognized as such because of one again, syntactic deviation, meaning they were mostly not disagreeing with the correct answers, but used different wording! (Joshi et al., 2017)

5 Generative QAS

Generative AI models continue to shake our cognition about how powerful AI tools can become over the last years, sparking major interest from researchers and everyday/casual users alike. Using Generative Question Answering, we can first-hand experience the SOTA level of human-like interaction with computers. With their high adaptability to vastly different types of questions posed by an estimated 100 million of visitors daily following

the GPT-4o announcement (Meer, 2024), Generative QAS proved that their understanding of human concepts will only get better and more refined. There are many reasons one might choose to build a GQAS, starting from their ability to efficiently utilize unlabeled data, learn interpretable dependencies among data, and incorporate an abundance of linguistic features without needing the extra effort of building a semantic parser, the heavy-weight language models come with an abundance of advantages.

5.1 General Architecture of Generative QAS

In this section, we will overview and examine the parts that make up a Generative Question Answering System:

The technical backbone of Generative Question Answering Systems begins with the aggregation and preparation of documents, where the data is parsed and chunked into digestible pieces (Miao, 2017). Given the user question represented as word sequence Q , the **Interpreter** encodes it to an array of vector representations. The interpreter utilizes transformer-based embeddings, converting the input question into these high-dimensional vector representations that encapsulates its semantic and syntactic properties.

This vector is then processed by the **Enquirer**, which fetches the relevant facts from a long-term memory knowledge base structured using memory networks or neural knowledge graphs. To retrieve the relevant information, the Enquirer can employ sparse retrieval mechanisms like TF-IDF and BM25, or dense retrieval mechanisms like BERT and Dense Passage Retrieval (DPR) to capture semantic similarity. The Enquirer utilizes attention mechanisms to prioritize the most significant parts of the knowledge base dynamically, and calculates the relevance (matching) scores between the question and the subject-relation-object triplets.

Taking the question embeddings and the long-term memory base into account, the **Answerer** combines an attention model with a transformer-based generative decoder to synthesize the final response. Specifically using multi-head attention models and sequence-to-sequence models like GPT-3 or T5, the Answerer is able to employ autoregressive language modeling to predict the subsequent tokens from scratch fluently.

5.2 Families of Generative Question-Answering Systems

Generative QAS leverage various methodologies to synthesize responses to user queries. In this section, we will explore some approaches that SOTA Large Language Models employ.

The Seq2Seq approach is a fundamental technique in generative QA systems, designed to map an input sequence to an output sequence (Ren et al., 2023). This architecture is fundamentally based on the Encoder and Decoder technologies. Encoder maps the query and context into a dense representation or context vector. Decoder is able to generate an output sequence (the answer) from the context vector, assigning the most probable word at each iteration (Lewis and Fan, 2018). Seq2Seq approaches are based on the Machine Translation problem. The general approach involves sentence generation by nature, as the translation in the target language has to be constructed from scratch. With the development of technologies such as LSTMs and GRU during the Deep Learning revolution, and the Transformer architecture, probabilistic modules where multiple Neural Network models combined into one giant network has become the staple in Generative NLP problems (Bahdanau et al., 2014).

Reinforcement Learning optimizes the answer generation process through feedback-based learning. The RL framework is characterized by the reward signal that scores the relevance, fluency and the coverage of answers from the user feedback. To briefly mention, RL models employ a policy network to generate candidate answers, and these answers are evaluated based on the reward signal. One SOTA research came up with a reward system especially for training QAS that focuses on the reformulation of questions. Since casual users of QAS may not favor sending manual feedback every time they get unsatisfying answers, the developers defined the reiteration or rewording of the original question as a negative feedback, and skipping to a new subject or continuing to explore the same topic with a different question as a positive feedback, signaling that the user was satisfied with the answer. To differentiate between these signals, the developers needed to employ powerful tools to detect paraphrasing (Kaiser et al., 2021). (Kaiser et.al, 2021)

5.3 Extractive Question Answering System

Example: ChatGPT

ChatGPT is a successor of the large language model (LLM) InstructGPT with a dialog interface that is fine-tuned using the Reinforcement Learning with Human Feedback (RLHF) approach. Developed by OpenAI (Metz, 2023), it is one of the most impressive NLP feats of today. In this section, we will take a closer look into the working mechanism of ChatGPT as a prime example of Generative Question Answering. ChatGPT comes arguably close to natural conversations due to its multi-prompt dialogue interface, and excels at human-like interactions. One of ChatGPT's shocking feats is its capacity to consider the entire conversation history, raising contextual awareness with every new query. Incorporating more human feedback with each new version due to its Reinforcement Learning from Human Feedback (RLHF) methodology, it can be argued that ChatGPT is becoming more capable of capturing humanistic nuances. Currently, it is able to outperform LLMs in zero-shot learning on various tasks in various mediums such as textual, visual or audio (Christiano et al., 2017).

To mention the technical qualities of ChatGPT, it uses multi-head self-attention to weigh the importance of different words to capture long-term dependencies as mentioned in the sections above, similarly to other Transformer based models. Moreover, ChatGPT is a Decoder-Only model, unlike traditional encoder-decoder models, predicting the next token based solely on previous tokens. This also shows parallelism with the objective function chosen to maximize the likelihood, namely, causal language modeling. ChatGPT is notably optimized with supervised fine-tuning: After pre-training, the model is fine-tuned on a narrower dataset with supervised learning, where human-labeled examples guide the model to generate more relevant and specific responses.

5.4 Performance of ChatGPT

In this section, a research (Bang et al., 2023) conducted to compare the performance of ChatGPT against other SOTA NLP technologies will be examined, aiming to understand how much has been achieved, and how much ground is yet to be covered with Generative Question Answering Systems.

Specifically designed to automate customer service chatbots, generating articles, blog posts, and

marketing content, or aiding people as a virtual personal assistant for scheduling, reminders, and information retrieval, ChatGPT is still lacking when it comes to some other NLP tasks. Researchers concluded that ChatGPT is the most powerful tool we have when it comes to Question-Answering tasks in terms of accuracy. However, ChatGPT is still prone to many mistakes when it comes to Machine Translation, and displays limited language understanding and generation capabilities on low-resource languages. In most tasks, especially task-oriented and knowledge-grounded dialogue tasks, task-specific fully-fine-tuned models outperform ChatGPT.

5.5 Challenges of LLM Based Question-Answering Systems

In this section we will discuss the drawbacks of LLM based QAS, explore the potential root causes of these challenges and determine which scenarios we should employ the help of these models.

One notable study (Bang et al., 2023) examined how ChatGPT performs when it comes to different reasoning types. To understand which ways ChatGPT is more successful, we will start by briefly defining the reasoning types mentioned in the study. Inductive reasoning involves drawing general conclusions from specific observations or instances. Induction requires figuring out new patterns and rules from a set of observations, and can be classified as a bottom-up strategy. For example, observing that the sun rises everyday from east, and concluding that it will always be so. On the other hand, deductive reasoning involves drawing specific conclusions from general principles or premises. It requires less guessing, and can be classified as a top-down strategy. The mentioned study tested the reasoning capabilities of ChatGPT and pointed out that it performs better deduction than induction. This result may come intuitive to readers as we discussed how LLMs exploit an extensive amount of information collected from a variety of different sources and incorporate intricate mechanisms to unveil the most relevant parts, narrowing the information in a top-down manner instead of coming up with the reasoning on its own. When it comes to other reasoning tasks, ChatGPT showed 63.41% accuracy on average in 10 different reasoning categories under logical reasoning, hence making it an unreliable reasoner. Specifically, it was pointed out that ChatGPT lacks simple mathematical and

Tasks	Dataset	Metric	Reference	Fine-Tuned SOTA	Zero-Shot SOTA	ChatGPT
Summarization	CNN/DM SAMSum	ROUGE-1 ROUGE-1	Lewis et al. (2020a)	44.47	35.27*	35.29
			Lewis et al. (2020a)	47.28	-	35.29
MT (XXX→Eng)	FLoRes-200 (HRL) FLoRes-200 (LRL)	ChrF++ ChrF++	Team et al. (2022)	63.5	-	58.64
			Team et al. (2022)	54.9	-	27.75
MT (Eng→XXX)	FLoRes-200 (HRL) FLoRes-200 (LRL)	ChrF++ ChrF++	Team et al. (2022)	54.4	-	51.12
			Team et al. (2022)	41.9	-	21.57
Sentiment Analysis	NusaX - Eng	Macro F1	Winata et al. (2022)	92.6	61.5	83.24
	NusaX - Ind	Macro F1	Winata et al. (2022)	91.6	59.3	82.13
	NusaX - Jav	Macro F1	Winata et al. (2022)	84.2	55.7	79.64
	NusaX - Bug	Macro F1	Winata et al. (2022)	70.0	55.9	55.84
Question Answering	bAbI task (15 16)	Accuracy	Weston et al. (2016a)	100 100	-	93.3 66.7
	EntailmentBank	Accuracy	Clark et al. (2018)	86.5	78.58	93.3
	CLUTRR	Accuracy	Minervini et al. (2020)	95.0	28.6	43.3
	StepGame (k=9 k=1)	Accuracy	Mirzaee and Kordjamshidi (2022)	48.4 98.7	-	23.3 63.3
	Pep-3k	AUC	Porada et al. (2021)	67.0	-	93.3
Misinformation Detection	COVID-Social	Accuracy	Lee et al. (2021)	77.7	50.0	73.3
	COVID-Scientific	Accuracy	Lee et al. (2021)	74.7	71.1	92.0
Task-Oriented Dialogue	MultiWOZ2.2	JGA	Zhao et al. (2022)	60.6	46.7	24.4
	MultiWOZ2.2	BLEU	Nekvinda and Dušek (2021)	19.1	-	5.65
	MultiWOZ2.2	Inform Rate	Yang et al. (2021)	95.7	-	71.1
Open-Domain KGD	OpenDialKG	BLEU ROUGE-L	Ji et al. (2022b)	20.8 40.0	3.1 29.5	4.1 18.6
	OpenDialKG	FeQA	Ji et al. (2022b)	48.0	23.0	15.0

Figure 1: Performance of ChatGPT compared to SOTA

spatial reasoning.

Another challenge pointed out by the researchers was the lack of common-sense reasoning. For instance, ChatGPT was prompted to judge if a predicate such as “knife chop rock” is plausible or implausible, which was deemed implausible by the language model. To the human eye, the prompted sentence carries some logic because of the popular “rock-paper-scissors” game, but to a language model, the information about how sturdy the mentioned materials concludes otherwise.

Generative QAS are also known to be susceptible to “model hallucinations” (Bang et al., 2023). Hallucinations are defined as the nonfactual, untruthful information generated by LLMs. Hallucinations can occur when the model has insufficient or incorrect information or overgeneralization, where the model predicts outputs based on patterns learned from the training data without grounding in factual accuracy or real-world knowledge. Two types of hallucinations are intrinsic, where the output contradicts the source/input content and extrinsic, where the output cannot be verified as neither supported nor contradicted by the source material. ChatGPT and other Generative QAS are known to be more likely to suffer from extrinsic hallucinations (Farquhar et al., 2024).

To list a few causes that result in hallucinations, misinterpretation of context, receiving the same

prompt multiple times, over-reliance on pre-trained knowledge can be interpreted as challenges LLMs are yet to overcome. However, the data circulating around the Web that LLMs refer to can also be ambiguous, unclear, biased, unstated or generally false.

6 Comparison of Extractive and Generative Question Answering Systems

So far, we discussed the architecture, workflow, and disadvantages of Extractive and Generative Question Answering Systems. In this section, we will compare two approaches and their optimal use cases.

Extractive models are faithful to their training data. This makes them more suitable for sensitive data applications where there is a high need for textual faithfulness, and lower tolerance for paraphrasing. They can also be utilized to fill out and populate knowledge graphs that can be used as training corpus by other models, increasing accuracy in a positive feedback loop. In summary, Extractive models are at the forefront when precision and accuracy are important. For instance, let us imagine a scenario in a legal context. When a question such as “What was the ruling in the 2020 Smith v. Jones case?” is prompted, an Extractive model will fetch the exact ruling from the case doc-

	In-domain Avg.			Out-of-domain Avg.		
	512	1024	Full	512	1024	Full
Single Task Learning						
T5 E-Ext (B)	74.42	75.80	77.93	55.89	58.06	58.65
T5 E-Ext (L)	76.46	78.67	80.85	60.74	63.67	64.49
T5 ED-Ext (B)	74.75	77.06	79.11	57.11	59.19	59.99
T5 ED-Gen (B)	77.91	80.68	81.02	56.26	61.75	61.82
BART E-Ext (L)	77.78	79.10	-	59.67	61.32	-
BART ED-Ext (L)	77.10	77.34	-	59.29	59.21	-
BART ED-Gen (L)	69.89	70.24	-	49.65	53.51	-
RoBERTa (L)	77.59	77.89	-	60.32	60.47	-
ELECTRA (L)	78.71	-	-	60.19	-	-
Multi-Task Learning						
T5 E-Ext (B)	75.74	76.65	78.99	58.94	61.55	61.98
T5 E-Ext (L)	77.10	79.30	81.55	63.04	66.10	66.78
T5 ED-Ext (B)	75.92	77.38	79.93	59.23	61.86	62.64
T5 ED-Gen (B)	78.06	80.89	81.16	57.82	63.56	63.68
BART E-Ext (L)	77.75	79.13	-	63.27	64.06	-
BART ED-Ext (L)	77.26	77.55	-	62.14	62.68	-
BART ED-Gen (L)	78.11	78.55	-	57.41	60.54	-
RoBERTa (L)	77.86	78.02	-	63.70	63.58	-
ELECTRA (L)	78.52	-	-	63.83	-	-

Figure 2: Comparison of Extractive vs. Generative readers (Luo et al., 2022)

ument, compared to a Generative model that will try to summarize or paraphrase the contents of the exact document, potentially losing legal nuances. A recent study published by Stanford showed that BERT-based models get around 76.5 F1, 73.2 EM on the test set; the final ensemble model gets 77.6 F1, 74.8 EM on the SQuAD 2.0 benchmark for Question Answering Systems.

Generative models are known for their natural and fluent creativity. This makes them the best candidate when the input context is not restricted, or is open-domain. Due to their large corpora, diverse vocabularies will be accepted by the language model. Generative models are a perfect fit for abstractive use cases such as translation, paraphrasing and summarization. One other aspect of its advantages is the conversational nature of the user interface, highlighting their capacity to conduct fluent and natural interactions. For instance, let us imagine a student that prompts the QAS with a complex concept, such as "Can you explain the significance of Einstein's theory of relativity?". In return, the Generative QAS produces a comprehensive explanation that compiles multiple aspects of the theory. In comparison, an Extractive model might pull disjointed sentences from various texts, and the end output can lack coherency. A 2024 study compared the performance of a language model that is based on GPT-3.5 and T5 on different datasets on the SQuAD benchmark. Results are shown in Figure 3 (Chowdhury and Chadha, 2023).

Dataset	SQUAD		NewWiki		NYT		Amazon		Reddit	
Metrics	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM
Real data	90.4	83.0	89.4	79.2	86.4	76.1	79.9	66.4	80.1	67.1
Generated data	79.5	64.6	80.1	65.3	76.5	63.2	72.4	59.5	72.7	60.2
Real + Wiki-samples	93.4	85.2	89.3	77.3	79.4	78.1	76.4	66.6	78.8	63.2
Real + Generated data	92.7	84.7	91.1	80.4	88.9	79.3	80.3	67.1	81.7	68.7

Figure 3: GPT-3.5 on SQuAD

Here, we can observe that a Generative model performs with 90.4 F1 and 83.0 EM on the SQuAD benchmark with real data. Generative models like GPT-3, used in ChatGPT, approach human performance levels in various QA tasks, achieving high scores on benchmarks such as TriviaQA, where it often scores above 80% accuracy in producing relevant answers.

Scholars (Luo et al., 2022) conducted a research comparing the success rate of Extractive and Generative readers, and subjecting them to questions of differing complexity and contexts. In Figure 1, we can observe that the Extractive readers perform better than the Generative reader in short context, but Generative ones perform better in long context. This result follows from the conclusions drawn in the previous segments, as Extractive models are more suited for close-domain contexts with high faithfulness to their source material, and Generative models are more suitable for open-domain general use cases.

7 Conclusion

In this paper, Question Answering Systems were examined in detail. Starting from their history in 1960s, QAS has come a long way, especially with the development of more advanced Machine Learning technologies. We can safely say that the introduction of more structured and linked knowledge bases on the Internet has aided tremendously to the information retrieval techniques that are fundamental in the training stage of QAS. We also examined two different types of QAS, being Extractive and Generative. The advantages, use cases and disadvantages of such systems are also inspected to understand in which cases these systems are optimal for use. Even though QAS technologies developed into irreplaceable tools in recent years, considering the evidence in this paper, it is important to con-

sider their shortcomings especially on complicated cognitive tasks and cases where accuracy of results is of utmost importance, such as medical or legal cases. QAS safety, reliability and explainability remain future work for the science of Question-Answering Systems.

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